Tomasz Kowalczyk

Senior Software Developer — Data Engineering

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PROFESSIONAL SUMMARY

Senior data engineer with 14 years of experience building data platforms for energy trading, retail, and banking. Designs batch and streaming pipelines with Apache Spark, Databricks, Kafka, dbt, and Airflow, and models data so analysts and data scientists can trust and reuse it. Led a lakehouse architecture that brought market and meter data freshness from 24 hours to 15 minutes while reducing compute cost by 31%. Introduced data contracts and automated data quality testing that cut monthly data incidents from 14 to 3. Mentors engineers and analysts on SQL, data modeling, and pipeline reliability, and is interested in cost-aware data architecture and streaming analytics.

TECHNICAL SKILLS

Programming languages: Advanced: SQL, Python, Scala. Working knowledge: Java, Bash, PL/SQL.

Data processing: Apache Spark (batch and Structured Streaming), Databricks, Delta Lake, Apache Kafka, Kafka Connect, Debezium change data capture. Working knowledge: Apache Flink.

Orchestration and transformation: Apache Airflow, dbt, Great Expectations. Working knowledge: Dagster.

Data warehouses and databases: Snowflake, Google BigQuery, PostgreSQL, Oracle, Apache Hive. Working knowledge: ClickHouse.

Data modeling and architecture: Lakehouse and medallion architecture, Kimball dimensional modeling, data contracts, schema registry (Avro, Protobuf), slowly changing dimensions, data vault basics.

Cloud and DevOps: Azure (Data Lake Storage, Event Hubs), Google Cloud (BigQuery, Dataflow), Terraform, Docker, GitLab CI.

Industry experience: Energy trading and smart metering, retail analytics, banking regulatory reporting.

Leadership and collaboration: Data platform guild lead, mentoring, cost reviews, collaboration with BI analysts, data scientists, and traders.

PROFESSIONAL EXPERIENCE

Halcyon Energy Trading (energy trading) — Warsaw, Poland

Senior Data Engineer — *Feb 2021 – Present*

- Architected a lakehouse on Databricks with Delta Lake and Spark Structured Streaming for smart meter readings and power market data (about 3 TB per day), replacing a nightly batch and bringing data freshness for trading analysts from 24 hours to 15 minutes.
- Introduced dbt models with Great Expectations tests and freshness checks across 220 tables, reducing data incidents reported by analysts from 14 to 3 per month.
- Reduced platform compute cost by 31% through cluster policies, autoscaling, Delta Z-ordering, partition pruning, and moving low-priority jobs to spot instances.
- Defined data contracts with upstream engineering teams using Avro schemas in a schema registry and CI compatibility checks, stopping unannounced schema changes from breaking downstream models.
- Lead the data platform guild and mentor four engineers; run monthly cost and reliability reviews with the head of data.

Pelikan Retail Group (retail) — Warsaw, Poland

Senior Data Engineer (Data Engineer until 2019) — *May 2016 – Jan 2021*

- Led the migration of an on-premises Hadoop and Hive warehouse to Google BigQuery, moving 400 tables and six years of history with reconciliation checks; most analyst reports went from hours to minutes.
- Built and operated about 600 Airflow DAGs for sales, inventory, and loyalty data, keeping a 99.5% on-time rate for morning reports used by 300 store managers.
- Implemented Kafka and Debezium change data capture from Oracle and PostgreSQL point-of-sale systems, giving the e-commerce team near-real-time stock visibility.
- Designed Kimball star schemas for sales and promotions with BI analysts, replacing 50 overlapping report-specific tables.

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Vistula Bank Services (banking IT services) Kraków, Poland
ETL Developer *Jul 2012 – Apr 2016*

- Developed SSIS packages and Oracle PL/SQL procedures for regulatory and management reporting at a mid-sized bank.
- Rewrote the nightly reporting batch with set-based SQL and table partitioning, cutting runtime from 9 hours to 3.5 hours.
- Built reconciliation reports between source systems and the reporting warehouse, used in monthly controls by the finance department.

SELECTED PROJECTS

dq-contracts — dbt package for data contracts (open source)
Technologies: dbt, SQL with Jinja, Python, GitHub Actions

- **Problem:** Teams documented expectations about source data in wikis that no pipeline enforced.
- **Contribution:** Wrote macros that generate dbt tests from YAML contracts and fail CI on breaking changes.
- **Result:** Adopted across Halcyon's nine data domains and by two teams in the Warsaw data community.

Streaming anomaly detection for smart meters
Technologies: Spark Structured Streaming, Delta Lake, Python, statistical control limits

- **Problem:** Faulty meters produced unrealistic readings that distorted demand forecasts.
- **Contribution:** Built a streaming job that flags readings outside rolling control limits and routes them to a quarantine table for review.
- **Result:** Detected 1,900 faulty meters in the first quarter and improved forecast input quality for the trading desk.

Data platform cost observability
Technologies: Databricks system tables, dbt, Power BI

- **Problem:** No team could see which pipelines drove compute cost.
- **Contribution:** Built cost attribution models by job, team, and table, with monthly reports for team leads.
- **Result:** Gave every team a clear view of its spend and directly supported the 31% cost reduction.

EDUCATION

Kraków University of Applied Informatics Kraków, Poland
M.Sc. Computer Science, specialization in Database Systems *2010 – 2012*

- Thesis: query optimization for partitioned fact tables in analytical databases.

Kraków University of Applied Informatics Kraków, Poland
B.Sc. Computer Science *2007 – 2010*

- Relevant coursework: database systems, distributed computing, statistics, software engineering.

HACKATHONS, CONFERENCES, AND PROFESSIONAL ACTIVITIES

- **Data Platform Days Warsaw (2025)** — Speaker: "Data contracts that engineers actually keep", lessons from rolling out contracts across nine data domains.
- **Open Data Hackathon Kraków (2019)** — Built an air-quality data pipeline and public dashboard from municipal sensor feeds with a team of four in 30 hours.
- **Warsaw Data Engineering Meetup (2020 – Present)** — Co-organizer of a quarterly meetup for data engineers and analysts.

CERTIFICATIONS AND ADDITIONAL QUALIFICATIONS

Certifications (illustrative): Lakehouse Architecture Professional Track, Northgate Data School (2022); Cloud Data Engineering Program, Meridian Cloud Academy (2018).

Mentoring: Mentors two junior data engineers each year through the Halcyon graduate program.

Languages: Polish (native), English (fluent), German (basic).

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